Check for updates

ACM DIGITAL LIBRARY

Association for Computing Machinery

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3746252.3761562

RESEARCH-ARTICLE

Towards Explainable Transaction Risk Analysis With Dual Graph Retrieval Augmented Generation

LIANG SU, Fudan University, Shanghai, China

MINGYANG ZHANG, Ant group, Hangzhou, Zhejiang, China

KANGXIANG JIA, Fudan University, Shanghai, China

TENGFEI LIU, Ant group, Hangzhou, Zhejiang, China

WEIQIANG WANG, Ant group, Hangzhou, Zhejiang, China

YUN XIONG, Fudan University, Shanghai, China

View all

Open Access Support provided by:

Fudan University

Ant group

University of California, Davis

PDF Download
3746252.3761562.pdf
12 February 2026
Total Citations: 0
Total Downloads: 107

Published: 10 November 2025

Citation in BibTeX format

CIKM '25: The 34th ACM International Conference on Information and Knowledge Management
November 10 - 14, 2025
Seoul, Republic of Korea

Conference Sponsors:
SIGWEB
SIGIR

Towards Explainable Transaction Risk Analysis With Dual Graph Retrieval Augmented Generation

Liang Su
Fudan University
Shanghai, China
lsu23@m.fudan.edu.cn

Mingyang Zhang
AntGroup
Hangzhou, China
zhangmingyang.zmy@antgroup.com

Kangxiang Jia
Fudan University
Shanghai, China
23212010013@m.fudan.edu.cn

Tengfei Liu
Weiqiang Wang
AntGroup
Hangzhou, China
aaron.ltf@antgroup.com
weiqiang.wwq@antgroup.com

Yun Xiong*
Xixi Wu
Fudan University
Shanghai, China
yunx@fudan.edu.cn
21210240043@m.fudan.edu.cn

Xinyu Gao
Yongrui Fu
Fudan University
Shanghai, China
23210240161@m.fudan.edu.cn
23210240154@m.fudan.edu.cn

Jiawei Zhang
University of California, Davis
Davis, California, USA
jiwzhang@ucdavis.edu

Abstract

Explainable transaction risk analysis is a challenge for traditional deep learning models, which only predict suspicious transactions without explanations. Current explainable methods rely on handcrafted rules and lack the ability to automatically generate language-based explanations. Large Language Models (LLMs) offer promise due to their reasoning and text generation abilities but struggle with domain knowledge and hallucinations, making risk analysis difficult. Specifically, LLMs face: **(1) insufficient adaptation to transaction data analysis**, and **(2) ineffective knowledge retrieval methods** that ignore the rich graph structure of transaction data. To address these issues, we propose the **Dual G**raph **R**etrieval-**A**ugmented **G**eneration (**Dual-gRAG**) framework, which utilizes dual retrieval: expert knowledge and reasoning case retrieval. Expert knowledge compensates for domain gaps, while reasoning case retrieval provides step-wise analysis guidance. We incorporate both graph-structured features and semantic features into the retrieval process to enhance the effectiveness of the retrieval. Extensive experiments show that Dual-gRAG improves LLMs' risk analysis capabilities, achieving a 15% increase in different metrics.

CCS Concepts

• **Information systems** → **Structured text search**; *Novelty in information retrieval*.

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
CIKM '25, Seoul, Republic of Korea
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-2040-6/2025/11
https://doi.org/10.1145/3746252.3761562

Keywords

Retrieval augmented Generation, Explainable financial risk analysis, Large language model

ACM Reference Format:
Liang Su, Mingyang Zhang, Kangxiang Jia, Tengfei Liu, Weiqiang Wang, Yun Xiong, Xixi Wu, Xinyu Gao, Yongrui Fu, and Jiawei Zhang. 2025. Towards Explainable Transaction Risk Analysis With Dual Graph Retrieval Augmented Generation. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), November 10–14, 2025, Seoul, Republic of Korea.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3746252.3761562

1 Introduction

In the financial domain, transaction risk analysis, particularly anti-money laundering (AML), is a critical billion-dollar challenge [3]. Traditional rule-based approaches, while offering clear and understandable decision-making processes, face significant drawbacks. They require extensive human labor and are increasingly inadequate due to the evolving techniques of money laundering and the growing volume of data [25]. Moreover, these rule-based systems can be easily bypassed by new or unknown laundering patterns [10]. As a result, deep learning technologies have gained recognition in AML detection. However, these models are often viewed as "black boxes," as they lack transparency and fail to provide explanations for their outcomes. This inability to generate natural language explanations significantly limits their usability and practical value, as users must rely on additional human effort to investigate and confirm the reasons behind the suspicion of a transaction [24].

Currently, researchers are striving to provide interpretability for deep learning methods. For instance, knowledge distillation techniques are used to extract knowledge from black-box models into transparent models, i.e., surrogate models, as a form of post-hoc model explanation [9, 35]. However, these methods generally fail to handle sequence data, i.e., transaction data. Other approaches

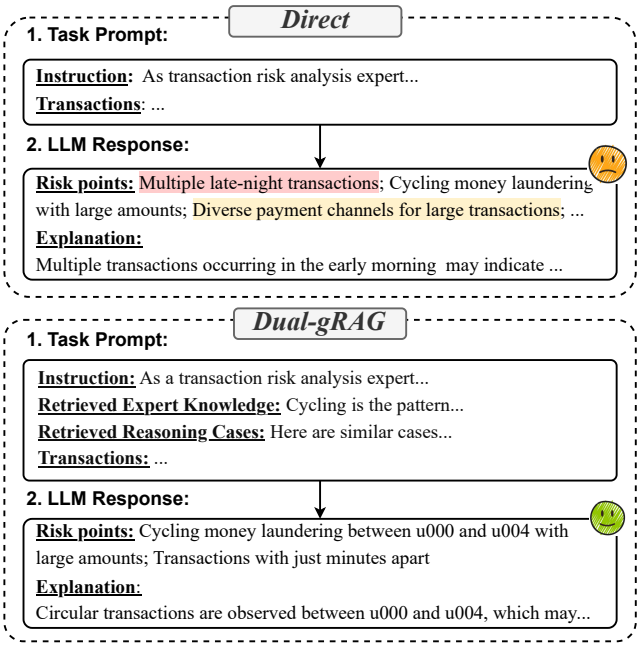

Figure 1: Dual-gRAG enables LLMs to perform explainable transaction risk analysis more effectively. In contrast, when LLMs analyze transaction risk directly, the results may contain incorrect conclusions and irrelevant findings.

[42] employ Monte Carlo Tree Search (MCTS) [8] to generate a set of statistics from the outputs of black-box models, subsequently composed into logical rules with neural logic network. Although such approach can distill sequence models into rule-based systems, it can only produce explanations based on pre-defined rules rather than intuitive language explanations.

Motivated by the growing capabilities of large language models (LLMs) in text generation, reasoning, and real-world applications [33], LLMs are increasingly seen as a promising tool for explainable AML solutions. One key advantage of LLMs in risk analysis is their ability to reason about new, previously unseen money laundering patterns, which can compensate for the limitations of traditional rule-based models. Their reasoning capabilities can capture emerging risks that rule-based systems may miss. Additionally, LLMs excel in text generation, enabling them to provide natural language explanations for flagged transactions, thus addressing the practical limitations of deep learning models in offering transparency. Recent studies [43] have demonstrated the strong performance of LLMs in natural language tasks, showcasing basic analytical and reasoning abilities. Preliminary research has also explored LLMs in financial tasks like sentiment analysis and stock prediction [23, 40]. However, their use specifically in AML remains limited.

Despite the remarkable performance of LLMs in language processing, they encounter significant challenges in tackling transaction risk analysis (example illustrated in Figure 1): **(1) Deficiency in transaction data analysis and reasoning.** LLMs are primarily designed for general natural language tasks, not structured transaction data [43]. This makes them ill-equipped to analyze and reason about laundering risk behaviors in transaction data. **(2) Unsuitable knowledge retrieval methods.** A challenge in using LLMs for risk analysis is their lack of domain-specific knowledge. While the RAG [27] algorithm was proposed to address this, many RAG methods rely on text-based similarity for retrieval [15]. These methods are inadequate for transaction data, which not only contains semantic features but also has a strong graph structure [28], making them unsuitable for risk analysis.

To address the aforementioned issues, we propose a novel framework, Dual Graph Retrieval Augmented Generation (**Dual-gRAG**). This framework features a dual retrieval approach from two augmented knowledge bases: an expert knowledge base and a reasoning case library. The expert knowledge base compensates for the lack of domain-specific knowledge, while the case library provides a step-by-step reasoning process for effective guidance. The retrieved reasoning cases serve as reference examples for LLMs, providing step-wise guidance to steer LLMs learn how to analyze transaction data and achieve accurate results. Relevant expert knowledge and similar representative cases are combined with input transaction records to enhance the reasoning capability of LLMs to generate reliable risk analysis. Specifically, we devise graph retriever that incorporate the semantics of transaction data as well as the structural patterns of transactions. By integrating the semantic and graph structure features of transaction data, we enhance the effectiveness of retrieval augmentation, thereby further improving LLMs' analytical capabilities. We conduct extensive experiments to demonstrate Dual-gRAG's effectiveness, showing an average improvement of 15% for all LLMs.

The contributions of this paper are summarized as follows:

- To the best of our knowledge, this is the first work leveraging LLMs for textual explainable transaction risk analysis. Unlike existing methods that only predict risk probability, our approach can generate textual analysis, providing a more applicable solution.
- We introduce a novel Dual-gRAG framework featuring dual graph retrievers with semantic and structure features of transaction data, addressing LLMs' inherent limitations.
- Extensive experiments show that our Dual-gRAG framework improves risk analysis performance by over 50% in precision and coverage across various LLMs.

2 Related Work

2.1 Retrieval Augmented Generation

RAG refers to a methodology that enhances model capability by integrating a retriever mechanism that accesses an external knowledge base [27]. When a query is submitted, the retriever identifies relevant documents from the knowledge base, which are then combined with the query to form the input for the model [7, 27, 36]. Subsequent advancements in this paradigm have introduced iterative retrieval [21, 34], self-reflective retrieval [4], and the integration of RAG with fine-tuning methodologies. Initially confined to NLP tasks, RAG has expanded to multi-modal settings, incorporating external knowledge such as code [17, 29] and images [11, 39, 41]. Some have even explored graph-based RAG, using knowledge graphs (KGs) as external base for retrival[14], or aligning query embeddings with code graph embeddings for retrieval [12].

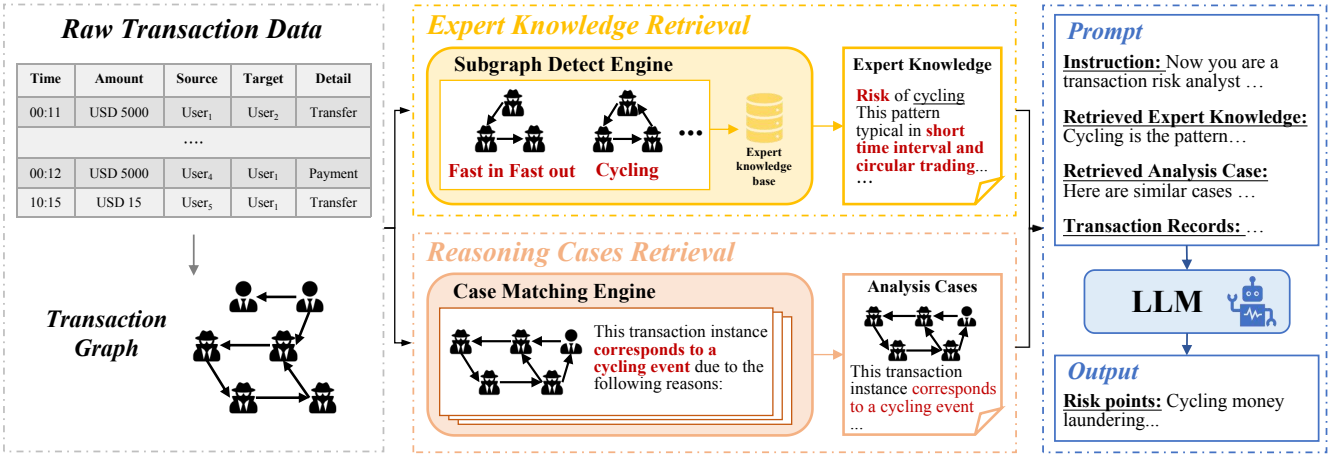

Figure 2: Overview of Dual-gRAG. It utilizes dual retrieval mechanisms, incorporating both relevant expert knowledge and similar reasoning cases to enable explainable prediction.

However, these methods are tailored for textual queries, which are transformed into text embeddings for retrieval [15]. This makes them unsuitable for risk analysis, where the input containing both textual and graph features, which our proposed method addresses by integrating a graph retriever.

2.2 Transaction Risk Analysis

Rule-based detection methods remain the mainstream choice in practice due to their clarity and ease of understanding [30]. However, they are easily bypassed by criminals, prompting research into supervised machine learning methods like support vector machines (SVM) [32] and decision trees (DT) [22]. While these methods can detect some risks, they struggle to capture new money laundering patterns [26].

Deep learning methods have been explored to address this limitation. Paule et al. [13] use Auto-Encoders for anomaly detection, while Han et al. [16] apply deep learning to enhance AML monitoring. Researchers also leverage the graph structure of transaction data for risk analysis, with Alarab et al. [2] and Weber et al. [31] using Graph Convolutional Networks (GCN) to detect illicit transactions.

In parallel, efforts have been made to develop interpretable risk analysis methods. Che et al. [9] employ knowledge distillation to explain black-box models, but struggle with sequential data. Zhang et al. [42] extracting logical rules from model outputs using MCTS. However, these methods still rely on manually defined rules, lacking intuitive textual explanations.

3 Methodology

In this section, we first formally define the task of explainable transaction risk analysis. Then, we present our framework (shown in Figure 2). The Dual-gRAG framework consists of two main components: **Expert Knowledge Retrieval**, which retrieves relevant expert knowledge, and **Reasoning Cases Retrieval**, which retrieves step-by-step analysis cases similar to the input transaction data. By utilizing two retrievers to acquire domain knowledge and step-wise analytical processes for risk analysis, we enhance the large model's risk analysis capabilities through chain-of-thought prompting.

3.1 Task Description

In the field of explainable transaction risk analysis, it is essential to analyze extensive transaction data to identify potential risk points. Traditional methods are limited to binary classification, indicating whether a transaction record is suspected of risk, i.e., True or False, without pinpointing the specific risk points.

The goal of this paper is to enable LLMs to analyze transaction data and uncover the hidden risk points. Formally, let $\mathcal{T} = \{T_1, T_2, \ldots, T_n\}$ represent the set of transaction records for a user within a specific timespan, where each transaction record T_i contains attributes such as time and amount. The input to the model is $\mathcal{T}$, and the output is a transaction risk analysis report $\mathcal{R} = \{'risk\ points' : P_1, P_2, \ldots, P_m;\ 'explanation' : E\}$, each phrase P_i corresponds to an estimated risk point, E corresponds to an intuitive textual explanation of the result. Detailed example refers to Figure 1.

3.2 Retrieval-augmented Framework

Our framework features a dual-retrieval approach, integrating both domain knowledge and reasoning cases. This significantly differs from existing RAG frameworks [27] that primarily address knowledge deficiencies. By incorporating step-by-step analysis details, the retrieved reasoning cases enhance LLMs' ability to evaluate transaction risks effectively. Moreover, both retrievers are graph-based, considering the semantic and structural features of transaction records, further distinguishing our method from current algorithms that rely solely on the semantic features of queries.

Since transaction data is inherently structured, we start by converting the input transaction data $\mathcal{T} = \{T_1, T_2, \ldots, T_n\}$ into a transaction graph $G(\mathcal{V}, \mathcal{E})$, as shown in the left part of Figure 2. Each

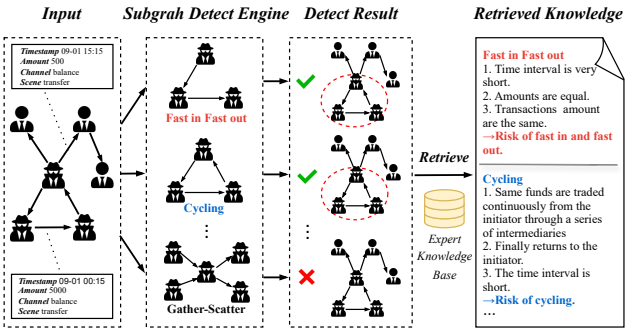

Figure 3: Illustration of Expert Knowledge Retrieval. Expert knowledge is encoded into graphs representing patterns. A detection engine determines if the input graph matches these predefined patterns.

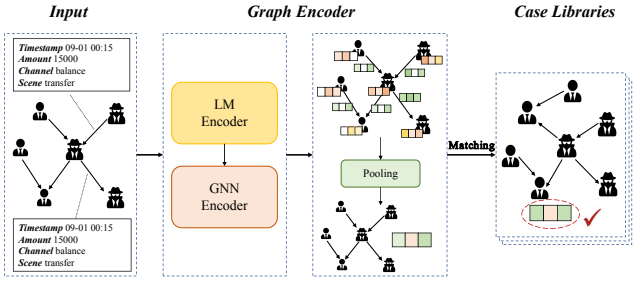

Figure 4: Illustration of Reasoning Cases Retrieval. We use an LM+GNN architecture to encode case and input graphs, retrieving the most similar cases based on embedding similarity.

transaction record $T_i \in \mathcal{T}$ represents an interaction between a source user u_i and a target user u_j. We define $\mathcal{V}$ as the set of all users involved, and $\mathcal{E} = \{(u_i, u_j)\}$ denotes the set of interactions between these users.

3.2.1 Expert Knowledge Retrieval. To build a high-quality knowledge base, we collect many analysis reports from experts and synthesize these into an expert knowledge base. Unlike other NLP tasks (such as QA), the knowledge in transaction risk analysis often exists in the form of "*Rules*". Therefore, the expert knowledge base $\mathcal{G}_{\text{know}}$ we constructed can also be referred to as a rule-base, with specific examples provided in the Section 6.1. Given that the knowledge is predominantly in the form of "*Rules*", our expert knowledge retrieval also adopts a detection engine trigger mechanism similar to that used in rule-based models, as illustrated in Figure 3.

Give an input transaction graph G derived from $\mathcal{T}$, expert knowledge retrieval follows a two-step process: First, subgraph matching is performed to detect if the input matches a known pattern graph $G_{\text{pattern}} \in \mathcal{G}_{\text{know}}$; then the attributes of the nodes and edges in the triggered pattern are examined for further verification. This approach ensures both structural and semantic accuracy by first matching the graph structure and then verifying feature similarity, such as interaction time and transaction amount.

Specifically, for "*Cycling*" subgraph matching, we first adopt Louvain algorithm [6] to derive a set of communities subgraphs of input as $\mathcal{S} = \{S_1, S_2, \ldots, S_m\}$ where each $S_i(\mathcal{V}_i, \mathcal{E}_i) \subseteq G$ denotes a subgraph and m is the total number of subgraphs. Then, we perform a one-by-one check between S_i and G_{pattern} to determine whether these two graphs are isomorphic, i.e., there exists a mapping $f : \mathcal{V}_i \rightarrow \mathcal{V}_{\text{pattern}}$, ensuring that $\forall (u, v) \in \mathcal{E}_i, (f(u), f(v)) \in \mathcal{E}_{\text{pattern}}$. This process yields a set of triggered patterns $\mathcal{G}' \subseteq \mathcal{G}_{\text{know}}$, representing the retrieved expert knowledge. As shown in Figure 3, parts of the input transaction graph match the *Fast-in-Fast-out* and *Cycling* patterns. We textualize these retrieved patterns to ensure LLMs can comprehend.

3.2.2 Reasoning Cases Retrieval. Similar to the expert knowledge, we construct a set of reasoning cases $\mathcal{G}_{\text{case}} = \{G_{c_1}, \ldots, G_{c_M}\}$, where each case G_{c_i} represents a transaction graph instance with a detailed textual analysis report t_{c_i}. These cases, collected from industrial platforms and verified by experts, ensure a diverse and high-quality set of reasoning processes. For detailed illustration please refer to Section 6.2. Unlike expert knowledge patterns with fixed structures, these cases exhibit diverse structures, making traditional graph searching algorithms inapplicable. Thus, we use neural retrieval techniques, encoding both input and case graphs into latent embeddings and matching via similarity, as shown in Figure 4.

Specifically, for the input graph G, we first leverage a pre-trained Language Model to generate initial node embeddings $\mathbf{X}$ for any $u \in \mathcal{V}$. As we can convert record T_i into plain text, an LM encoder can be leveraged to generate T_i's representations as $\mathbf{t}_i = \text{LM}(T_i)$, then a node's initial embedding is computed as mean-pooling of related transactional edges as $\mathbf{x}_u = \text{Mean-Pooling}(\{\mathbf{t}_i | u \in \text{source/target users of } T_i\})$. Then, the GNN further captures the structural patterns of this transaction graph, leading to refined node embedding matrix as $\mathbf{H} = \text{GNN}(\mathbf{X}, \mathcal{E})$. We further use a Read-Out function, i.e., Sum-Pooling, to obtain this transcation graph's embedding as $\mathbf{h}_G = \text{Sum-Pooling}(\mathbf{H})$. As all case graphs in the case library can be encoded in the same way, the case library can be represented as $\mathbf{H}_{\text{case}} = \{\mathbf{h}_{G_{c_1}}, \ldots, \mathbf{h}_{G_{c_M}}\}$ where M refers to the number of available cases and $\mathbf{h}_{G_{c_i}}$ denotes the representation of the i-th case graph G_{c_i}. We select the top-K most similar cases from the case library to consisitue the retrival results, as $C' = \text{Top-K}_{G_{c_i} \in \mathcal{G}_{\text{case}}} \text{Similarity}(\mathbf{h}_G, \mathbf{h}_{G_{c_i}})$.

This method ensures retrieval of the most contextually relevant cases by considering **both the semantic and structural features** of the transaction records. Note that such retrieval can be **training-free** as it leverages a pre-trained LM and a no-parameter GNN, i.e., LightGCN [18]. The retrieved reasoning cases are incorporated into the prompts as few-shot-CoT examples for the LLMs, steer them to perform chain-of-thought reasoning, thus enhancing their inferential capabilities [38].

3.2.3 Input Construction. After completing the dual-retrieval process, we combine three key elements to create comprehensive inputs for LLMs: (1) Expert knowledge serves as the foundational background for model inference. (2) Reasoning cases act as few-shot learning examples to assist the model in analysis. (3) The input transaction data is converted into plain text for the model to analyze.

4 Experiments

To facilitate explainable transaction risk analysis, we construct a new transaction benchmark dataset: **RA-bench**. Distinct from other anti-money laundering datasets, our dataset not only includes comprehensive labels but also annotates the specific money laundering patterns involved in anomalous transactions. Our work aims to answer the following three research questions:

- **RQ1:** How much improvement can the Dual-gRAG framework bring to LLMs?
- **RQ2:** How do the two proposed retrieval components contribute to the effectiveness of Dual-gRAG?
- **RQ3:** How does the proposed graph retrieval method perform against other retrieval methods?

4.1 Experimental Settings

4.1.1 Baselines. To demonstrate the effectiveness of the Dual-gRAG framework, we employ several traditional machine learning models as our baselines and classic LLMs as backbone models.

Traditional Machine Learning Baselines:

- **LightGCN+SVM**: This model uses LightGCN to capture the structural information of the transaction graph, followed by pooling node embeddings for graph-level representation, which is then classified by an SVM.
- **LightGCN+MLP**: This model replaces the SVM with an MLP to explore deep learning performance in this task.
- **GCN+MLP**: Here, a trainable GCN replaces LightGCN, and the GCN and MLP are co-trained in an end-to-end framework for classification.

For all methods, transaction data is converted into a transaction graph, and node features are extracted by language model as described in Section 3.2.2.

LLM Backbones:

- **Llama3-8B-Instruct** [1], a powerful large language model introduced by Meta.
- **Qwen2.5-14B-Instruct** [5], a transformer-based decoder-only language model.
- **Qwen2.5-72B-AWQ** [5], a 72B large language model quantized with AWQ.

For all LLMs, we provide the same prompts to eliminate performance deviations caused by prompt differences.

4.1.2 Datasets. To better evaluate the effectiveness of Dual-gRAG, we conduct experiments on two datasets, namely:

- **RA-Bench**: an open-source dataset based on real-world scenarios that we constructed to facilitate further research by the community.[1]
- **IT-AML**[3]: an open-source AML dataset released by IBM.

4.1.3 Evaluation Metrics. We use GPT-4[2] to evaluate the results, mitigating the influence of human emotional bias [23]. The evaluation process consists of two parts: (1) comparing the risk points identified by the model with those in the ground truth labels, and (2) assessing the quality of the explanations generated by the model.

[1]The data is available on https://github.com/V1ncent-S/RA-Bench.
[2]All experimental data has been strictly anonymized, ensuring no risk of privacy leakage.

To provide a multi-dimensional view of the LLM's risk analysis capabilities, we employ three evaluation metrics. Additionally, we introduce another metric specifically for machine learning baselines to compare the performance of these methods with that of LLMs in the risk analysis task. The specific evaluation metrics are as follows:

- **Coverage:** Evaluate how many of the risk points identified by the model overlap with the true risk points.
- **Precision:** Evaluate how many of the risk points identified by the model are actually true risk points.
- **ROUGE:** Evaluate the quality of the explanations generated by the model, we use ROUGE-L as the evaluation metric.
- **Accuracy:** Evaluate whether deep learning models or LLMs can correctly identify whether the transaction record graph indicates potential risks.

4.1.4 Implementation Details. For reasoning case retrieval, we employ m3e [37] (an open-source massive mixed embedding model) as our language model encoder, top-K is 1. Furthermore, to investigate the improvements that RAG combined with fine-tuning can bring to explainable risk analysis tasks, we utilize the Dual-gRAG framework to construct an instruction fine-tuning dataset. Subsequently, we use LoRA[19] to fine-tune the Qwen2.5-14B on one A100—80G GPUs with the learning rate of 1e-4 to develop an LLM specifically focused on risk analysis task: **RA-GPT** (Risk Analysis GPT).

4.2 Performance Comparison(RQ1)

For overall performance comparison, we provide both quantitative results (Table 1) and qualitative results (Figure 1).

4.2.1 Quantitative Results. In this section, we quantitatively evaluate the effectiveness of the Dual-gRAG framework on explainable risk analysis tasks. Table 1 reports results across four key metrics relevant to risk analysis.

For the first two evaluation metrics, we observe that nearly all LLMs outperform direct input analysis under the Dual-gRAG framework, with an average improvement of around 15%. This suggests that Dual-gRAG helps reduce irrelevant or incorrect outputs, thereby improving the coverage and precision of the generated analysis.

For ROUGE scores—which measure the quality of generated explanations—we observe significant gains as well. All models show nearly a 50% increase in ROUGE scores, indicating that Dual-gRAG substantially enhances the clarity and relevance of the reasoning process and the reports. These improvements lead to more accurate and concise explanations, thus increasing the overall utility of the models.

In the IT-AML dataset, multi-currency transactions require historical exchange rates, which are unavailable. This leads to inconsistencies in transaction amounts within the input networks, differing from the expert knowledge that assumes similar amounts across money laundering patterns. As a result, the model is constrained by these assumptions, reducing its ability to detect varied risk patterns and limiting coverage. In contrast, in the direct scenario, the model analyzes suspicious flows more freely without being restricted by amount similarity. Nonetheless, Dual-gRAG enhances precision by leveraging expert knowledge and historical cases, effectively reducing irrelevant or incorrect outputs.

Table 1: Overall Performance Comparison (RQ1). "Direct" refers to LLM's direct inference (example illustrated in Figure 1), and "Dual-gRAG" denotes the integration with our framework, which can consistently bring significant improvement on all metrics.

Dataset	Model	Coverage %		Precision %		ROUGE %	
		Direct	Dual-gRAG	Direct	Dual-gRAG	Direct	Dual-gRAG
RA-Bench	Qwen2.5-14B	51.64	**53.28**	45.32	**72.63**	23.67	**34.92**
	Llama3-8B	44.88	**57.79**	48.67	**55.08**	27.54	**32.52**
	RA-GPT	47.95	**62.09**	44.32	**66.45**	25.41	**42.14**
	Qwen2.5-72B	51.84	**84.22**	49.03	**82.86**	26.87	**47.89**
IT-AML	Qwen2.5-14B	**72.88**	62.78	54.27	**80.66**	21.98	**26.38**
	Llama3-8B	59.38	**61.93**	51.73	**54.23**	18.16	**19.35**
	RA-GPT	**34.01**	26.64	36.56	**41.94**	22.32	**25.42**
	Qwen2.5-72B	75.85	**91.47**	57.29	**80.90**	25.54	**43.64**
Real-World	Qwen2.5-14B	34.51	**48.98**	34.71	**43.87**	18.24	**19.20**
	Llama3-8B	48.48	**53.30**	41.52	**45.45**	20.05	**20.69**
	RA-GPT	40.38	**44.23**	30.43	**41.07**	17.13	**21.91**
	Qwen2.5-72B	58.88	**85.79**	40.27	**76.02**	22.92	**46.35**

Model	$SVM_{LightGCN}$	$MLP_{LightGCN}$	GCN	Qwen2.5-14B	RA-GPT
Accuracy	0.65	0.70	0.72	0.93	0.97

Additionally, since traditional machine learning models (non-LLMs) cannot generate textual explanations, we only report their performance on money laundering pattern classification. Results show that, under the Dual-gRAG framework, LLMs significantly outperform other deep learning models. This advantage likely stems from LLMs' ability to explicitly learn knowledge about different laundering patterns and apply reasoning guided by examples to detect suspicious behavior in transaction graphs—leading to improved analytical accuracy.

4.2.2 Case Study. In addition to quantitative metrics, we explore how the Dual-gRAG framework enhances the model's performance in risk analysis. For this, we select an example from Qwen2.5's output to demonstrate the improvement, as shown in Figure 1.

From Figure 1, we see that while the LLM correctly identifies risk points in the transaction data, it also generates incorrect (red) and irrelevant (orange) content. With the Dual-gRAG framework, the LLM's analysis becomes more precise, likely due to the retrieved cases that helps the model focus on the provided knowledge. The framework not only improves accuracy but also identifies specific money laundering users, such as the cycle between u000 and u004 (see Figure 1). The step-by-step reasoning rationale in the cases guides the LLM to analyze potential risks thoroughly, resulting in more detailed outcomes.

In contrast to traditional machine learning models with binary outputs, our results provide detailed explanations for suspected transaction risks, offering high interpretability. Thus, staff can easily verify the identified risks during the validation phase.

4.2.3 Online Test. We further deploy Dual-gRAG in a real-world risk analysis scenario to evaluate its practical utility. As shown in Table 1, Dual-gRAG consistently improved LLM performance—achieving a nearly 25% gain in coverage, over 30% in precision, and a 5% increase in ROUGE—demonstrating its effectiveness in applications.

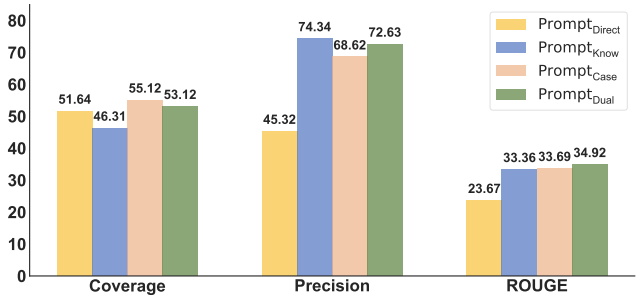

Figure 5: Ablation study of Dual-gRAG with Qwen2.5-14B serving as the base LLM.

Moreover, the risk analysis reports generated by the LLM with Dual-gRAG significantly enhance analyst efficiency. The high interpretability of the outputs allows staff to easily verify potential money laundering cases and take appropriate enforcement actions.

4.3 Ablation Study (RQ2)

To assess the effectiveness of the Dual-gRAG framework, we conduct an ablation experiments to validate the contribution of each retrieval component. The results are shown in Figure 5.

We test different prompting methods, incorporating expert knowledge and reasoning cases, to evaluate performance improvements.

The four prompting methods are: Prompt_{Direct} (direct risk analysis), Prompt_{Know} (expert knowledge only), Prompt_{Case} (reasoning cases only), and Prompt_{Dual}.

From Figure 5, we observe that Prompt_{Case} performs well. While Prompt_{Know} shows good precision, it has lower coverage than Prompt_{Direct}, likely because expert knowledge reduces incorrect risk points, improving precision. However, using only expert knowledge limits the model's inherent knowledge, leading to reduced diversity in the identified risk points, reducing coverage.

Prompt_{Case}, by providing detailed case guidance, allows the LLM to infer more potential risk points. However, due to the lack of corresponding expert knowledge, the precision may decrease. In contrast, Prompt_{Dual}, through the dual retrieval method, achieves a good balance between coverage and precision.

4.4 Retrieval Method Comparison (RQ3)

In this section, we examine the effectiveness of the graph retriever within the Dual-gRAG framework. Due to the low semantic similarity between the input transaction data and expert knowledge, text similarity cannot be used for expert knowledge retrieval. Thus, we use simple semantic information (i.e., text similarity between transaction in the input and case) for reasoning case retrieval. Given that both the input and knowledge base in our task scenario exhibit graph-structured features, we also use the GRAG [20] algorithm as a baseline retrieval method. The results are shown in Table 2.

From Table 2, we see that LLM performance with Simple-RAG decreases across all metrics compared to Dual-gRAG. This indicates that relying solely on text similarity reduces retrieval quality and impacts risk analysis capabilities. Additionally, we observe that the performance of GRAG is suboptimal, likely because the GRAG method does not adapt well to our task scenario. Its pruning operation based on semantic information may lead to the loss of key details (e.g., the removal of a transaction related to a crime), which results in degraded retrieval performance.

Table 2: Comparison of Graph-Retrieval with Qwen2.5-14B serving as the base LLM.

Method	Coverage (%)	Precision (%)	ROUGE (%)
Direct	51.64	45.32	23.67
Simple-RAG	46.93	60.58	21.19
GRAG	45.70	57.77	22.36
Dual-gRAG	**53.28**	**72.63**	**34.92**

5 Efficiency Discussion

In financial transaction risk analysis, model efficiency is critical. Our framework maintains low complexity: expert knowledge retrieval uses a traditional subgraph matching approach with complexity linear to the number of patterns, and case retrieval employs LightGCN with complexity linear to the number of edges. Since each graph is small, the overall retrieval complexity is $O(\mathcal{V} + \mathcal{E})$.

Additionally, we focus on analyzing a user's 1 or 2-hop transaction network within a single day, keeping the graph size manageable and ensuring fast retrieval. When used by staff, the LLM can directly present explainable user-level transaction risks, requiring only minimal validation and significantly reducing manual workload.

6 Retrieval Knowledge Base

6.1 Expert Knowledge Base

From the collected analysis reports, we extract key expert knowledge for risk analysis, including textual descriptions of four known money laundering patterns and three additional common risk indicators (e.g., frequent early-morning transactions, sudden large transfers). As expert knowledge in transaction risk analysis is often rule-based, our knowledge base resembles a rule base. However, rule-based models are prone to being bypassed. To address this, we incorporate general risk indicators alongside typical patterns, leveraging LLM reasoning to identify emerging risks and mitigate the limitations of fixed rules. All seven risk points are manually verified and integrated into our expert knowledge base.

For example, the "*Cycling*" pattern is a high-risk transaction mode where funds flow through several users and eventually return to the original sender, forming a closed loop. This cycle pattern is represented as a graph G_{cycle} with a node set representing distinct users $\mathcal{V}_{\text{cycle}} = \{v_1, ..., v_k\}$, and edges forming a cycle $\mathcal{E}_{\text{cycle}} = \{(v_1, v_2), \ldots, (v_k, v_1)\}$. Similarly, we can encode a variety of transaction patterns in graph format, resulting in a **graph-formatted expert knowledge base** $\mathcal{G}_{\text{know}} = \{G_{\text{cycle}}, G_{\text{Fast-in-Fast-out}}, G_{\text{Scatter}}, \ldots\}$, where each G_{pattern} represents a graph encoding a distinct pattern of a high-risk laundering mode. Illustrative examples of these patterns are shown in Figure 3.

6.2 Reasoning Case Library

After aggregating a substantial number of historical analysis cases from industrial platforms, we meticulously filter and select representative cases that encompass diverse combinations of risk indicators to ensure the comprehensive diversity of the case library. Subsequently, domain experts rigorously examine these cases to validate the accuracy of the analysis results and processes, further refining the analytical procedures embedded within each case. Through extensive and systematic validation, we guarantee that the case library is not only diverse but also of exceptional quality.

7 Conclusion

In this study, we explore the task of explainable transaction risk analysis. We identify two key challenges for this task: (1) Deficiency in transaction data analysis and reasoning, and (2) Incompatibility of existing RAG methods with risk analysis scenarios. To tackle these issues, we propose the Dual-gRAG framework, which employs dual-retrieval thought and graph retrieval techniques to provide LLMs with the expert knowledge and reasoning cases necessary for risk analysis. Extensive experiments and online tests demonstrate that Dual-gRAG significantly enhances the risk analysis capabilities of LLMs and its strong applicability in real scenarios.

Acknowledgments

This work was supported by Ant Group.

GenAI Usage Disclosure

In our experiments, GPT-4 was used to evaluate the coverage and accuracy of outputs generated by different models. We also validated the effectiveness of the Dual-gRAG method using various LLMs, as detailed in Section 4.1.1. The LLM-generated examples shown (e.g., Figure 1) are used to illustrate experimental outcomes and demonstrate the improvements achieved by Dual-gRAG.

References

[1] AI@Meta. 2024. Llama 3 Model Card. (2024). https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md

[2] Ismail Alarab, Simant Prakoonwit, and Mohamed Ikbal Nacer. 2020. Competence of Graph Convolutional Networks for Anti-Money Laundering in Bitcoin Blockchain. *Proceedings of the 2020 5th International Conference on Machine Learning Technologies* (2020).

[3] Erik Altman, Jovan Blanuša, and Luc Von Niederhäusern et al. 2023. Realistic Synthetic Financial Transactions for Anti-Money Laundering Models. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

[4] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *ArXiv* abs/2310.11511 (2023).

[5] Jinze Bai, Shuai Bai, and Yunfei Chu et al. 2023. Qwen Technical Report. *arXiv preprint arXiv:2309.16609* (2023).

[6] Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment* 2008 (2008), P10008.

[7] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, and Trevor Cai et al. 2021. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning*.

[8] Cameron Browne, Edward Jack Powley, Daniel Whitehouse, Simon M. M. Lucas, Peter I. Cowling, Philipp Rohlfshagen, Stephen Tavener, Diego Perez Liebana, Spyridon Samothrakis, and Simon Colton. 2012. A Survey of Monte Carlo Tree Search Methods. *IEEE Transactions on Computational Intelligence and AI in Games* 4 (2012), 1–43.

[9] Zhengping Che, S. Purushotham, Robinder G. Khemani, and Yan Liu. 2015. Distilling Knowledge from Deep Networks with Applications to Healthcare Domain. *ArXiv* abs/1512.03542 (2015).

[10] Tianyi Chen and Charalampos Tsourakakis. 2022. AntiBenford Subgraphs: Unsupervised Anomaly Detection in Financial Networks. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*. Association for Computing Machinery. doi:10.1145/3534678.3539100

[11] Wenhu Chen, Hexiang Hu, and Xi Chen et al. 2022. MuRAG: Multimodal Retrieval-Augmented Generator for Open Question Answering over Images and Text. In *Conference on Empirical Methods in Natural Language Processing*.

[12] Kounianhua Du, Renting Rui, Huacan Chai, Lingyue Fu, Wei Xia, Yasheng Wang, Ruiming Tang, Yong Yu, and Weinan Zhang. 2024. CodeGRAG: Extracting Composed Syntax Graphs for Retrieval Augmented Cross-Lingual Code Generation. *ArXiv* abs/2405.02355 (2024).

[13] LPaula Ebberth, Ladeira Marcelo, NCarvalho Rommel, and Marzagao Thiago. 2016. Deep Learning Anomaly Detection as Support Fraud Investigation in Brazilian Exports and Anti-Money Laundering. (2016).

[14] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. *ArXiv* abs/2404.16130 (2024).

[15] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Qianyu Guo, Meng Wang, and Haofen Wang. 2023. Retrieval-Augmented Generation for Large Language Models: A Survey. *ArXiv* abs/2312.10997 (2023).

[16] Jingguang Han, Utsab Barman, Jeremiah Hayes, Jinhua Du, Edward Burgin, and Dadong Wan. 2018. NextGen AML: Distributed Deep Learning based Language Technologies to Augment Anti Money Laundering Investigation. In *Annual Meeting of the Association for Computational Linguistics*.

[17] Shirley Anugrah Hayati, Raphaël Olivier, Pravalika Avvaru, Pengcheng Yin, Anthony Tomasic, and Graham Neubig. 2018. Retrieval-Based Neural Code Generation. *ArXiv* abs/1808.10025 (2018).

[18] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (2020).

[19] J. Edward Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *ArXiv* abs/2106.09685 (2021).

[20] Yuntong Hu, Zhihan Lei, Zhengwu Zhang, Bo Pan, Chen Ling, and Liang Zhao. 2024. GRAG: Graph Retrieval-Augmented Generation. In *North American Chapter of the Association for Computational Linguistics*. https://api.semanticscholar.org/CorpusID:270062608

[21] Zhengbao Jiang, Frank Xu, and Luyu et al. Gao. 2023. Active Retrieval Augmented Generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 7969–7992. doi:10.18653/v1/2023.emnlp-main.495

[22] Martin Jullum, Anders Løland, Ragnar Bang Huseby, Geir Ånonsen, and Johannes Lorentzen. 2020. Detecting money laundering transactions with machine learning. *Journal of Money Laundering Control* 23, 1 (2020).

[23] Kelvin J.L. Koa, Yunshan Ma, Ritchie Ng, and Tat-Seng Chua. 2024. Learning to Generate Explainable Stock Predictions using Self-Reflective Large Language Models. *Proceedings of the ACM on Web Conference 2024* (2024).

[24] Dattatray Vishnu Kute, Biswajeet Pradhan, Nagesh Shukla, and Abdullah Alamri. 2021. Deep Learning and Explainable Artificial Intelligence Techniques Applied for Detecting Money Laundering–A Critical Review. *IEEE Access* 9 (2021), 82300–82317. doi:10.1109/ACCESS.2021.3086230

[25] Nevine Makram Labib, Mohammed Abo Rizka, and Amr Ehab Muhammed Shokry. 2020. Survey of machine learning approaches of anti-money laundering techniques to counter terrorism finance. In *Internet of Things—Applications and Future: Proceedings of ITAF 2019*. Springer, 73–87.

[26] Nevine Makram Labib, Mohammed Abo Rizka, and Amr Ehab Muhammed Shokry. 2020. Survey of Machine Learning Approaches of Anti-money Laundering Techniques to Counter Terrorism Finance.

[27] Patrick Lewis, Ethan Perez, and Piktus et al. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS '20)*. Curran Associates Inc., Red Hook, NY, USA, Article 793, 16 pages.

[28] Xujia Li, Yuan Li, Xueying Mo, Hebing Xiao, Yanyan Shen, and Lei Chen. 2023. Diga: Guided Diffusion Model for Graph Recovery in Anti-Money Laundering. *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (2023).

[29] Mingwei Liu, Tianyong Yang, Yiling Lou, Xueying Du, Ying Wang, and Xin Peng. 2023. CodeGen4Libs: A Two-Stage Approach for Library-Oriented Code Generation. *2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE)* (2023), 434–445.

[30] Marvin Oeben, JeroenP. Goudsmit, and Elena Marchiori. 2019. Prerequisites and AI challenges for model-based Anti-Money Laundering. (Aug 2019).

[31] Aldo Pareja, Giacomo Domeniconi, Jie Chen, Tengfei Ma, Toyotaro Suzumura, Hiroki Kanezashi, Tim Kaler, and Charles E. Leisersen. 2019. EvolveGCN: Evolving Graph Convolutional Networks for Dynamic Graphs. *ArXiv* abs/1902.10191 (2019).

[32] Omri Raiter. 2021. Applying supervised machine learning algorithms for fraud detection in anti-money laundering. *Journal of Modern Issues in Business Research* 1, 1 (2021), 14–26.

[33] Wenbo Shang and Xin Huang. 2024. A Survey of Large Language Models on Generative Graph Analytics: Query, Learning, and Applications. arXiv:2404.14809 [cs.CL]

[34] Zhihong Shao, Yeyun Gong, and Yelong et al. Shen. [n. d.]. Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 9248–9274. doi:10.18653/v1/2023.findings-emnlp.620

[35] S. Tan, Rich Caruana, Giles Hooker, and Yin Lou. 2017. Distill-and-Compare: Auditing Black-Box Models Using Transparent Model Distillation. *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (2017).

[36] Boxin Wang, Wei Ping, Lawrence C. McAfee, Peng Xu, Bo Li, Mohammad Shoeybi, and Bryan Catanzaro. 2023. InstructRetro: Instruction Tuning post Retrieval-Augmented Pretraining. *ArXiv* abs/2310.07713 (2023).

[37] He sicheng Wang Yuxin, Sun Qingxuan. 2023. *M3E: Moka Massive Mixed Embedding Model*.

[38] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Huai hsin Chi, F. Xia, Quoc Le, and Denny Zhou. 2022. Chain of Thought Prompting Elicits Reasoning in Large Language Models. *ArXiv* abs/2201.11903 (2022).

[39] Chen-Wei Xie, Siyang Sun, Xiong Xiong, Yun Zheng, Deli Zhao, and Jingren Zhou. 2023. RA-CLIP: Retrieval Augmented Contrastive Language-Image Pre-Training. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2023), 19265–19274.

[40] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-Source Financial Large Language Models. arXiv:2306.06031 [q-fin.ST]

[41] Michihiro Yasunaga, Armen Aghajanyan, Weijia Shi, Rich James, Jure Leskovec, Percy Liang, Mike Lewis, Luke Zettlemoyer, and Wen tau Yih. 2023. Retrieval-Augmented Multimodal Language Modeling. *ArXiv* abs/2211.12561 (2023).

[42] Yao Zhang, Yun Xiong, Yiheng Sun, Caihua Shan, Tian Lu, Hui Song, and Yangyong Zhu. 2022. RuDi: Explaining Behavior Sequence Models by Automatic Statistics Generation and Rule Distillation. *Proceedings of the 31st ACM International Conference on Information & Knowledge Management* (2022).

[43] Wayne Xin Zhao, Kun Zhou, and Junyi Li et al. 2023. A Survey of Large Language Models. arXiv:2303.18223 [cs.CL]